

Front

60 mm

60 mm

240 mm

For Hospital or Medical practitioner or Laboratory

CONTRAPAQUE 300

CONTRAPAQUE 350

(Iohexol Injection USP)

Name and strength of Active ingredients for Contrapaque 300 mg:

Each ml contains:

Iohexol USP647 mg

Equivalent to Iodine300 mg

Water for injection BPq.s

Name and strength of Active ingredients for Contrapaque 350mg:

Each ml contains:

Iohexol USP755 mg

Equivalent to Iodine350 mg

Water for injection BPq.s

Description:

Clear, colourless to pale yellow slightly viscous liquid, free from foreign matter.

Pharmacodynamics:

Iohexol is a "second generation nonionic radio contrast agent. It is available for intravenous, intra-arterial and intra-thecal administration in diagnostic radiology. In comparison with conventional (ionic) radiocntrast media (eg, metrizoate, iothalamte and diatrizoate) iohexol has a relatively low osmolality. This lower osmolality has been associated with a lower incidence of side effects such as pain, heat sensations, adverse hemodynamic changes, EKG changes, endothelial damage and damage to erythrocytes.

Pharmacokinetics:

Absorption

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Following intrathecal injection, iohexol is rapidly absorbed into the bloodstream from CSF. Peak plasma concentrations occur between 2 and 6 hours after injection.

Distribution

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Iohexol has low affinity for serum or plasma proteins and is poorly bound to serum albumin.

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After intravascular injection iohexol is distributed in the extra cellular fluid compartment, with no significant deposition. Iohexol does not cross the blood- brain barrier to any significant extent following intravascular administration. Iohexol probably crosses the placental barrier in humans by simple diffusion. The volume of distribution is reported to be 0.577 L/kg (0.35 to 0.849 L/kg) after lumbar intrathecal injection.

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After intravenous injection, the volume of distribution has been reported to be 0.165 L/kg (0.108 to 0.219 L/kg)

Metabolism

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Iohexol undergoes minimal metabolism, de-iodination, or bio-transformation. No metabolites of iohexol have been detected in urine, feces or plasma.

Excretion

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Following intrathecal or intravenous administration, iohexol is excreted primarily through the kidneys, almost entirely via glomerular filtration. In patients with severe renal insufficiency or failure, compensatory biliary excretion of the drug is anticipated to occur, with a slow clearance into the bile. It is not known to what extent iohexol is excreted in human milk.

Half-Life

The elimination half-life to iohexol, after intravenous administration, is approximately 2 hours. After intrathecal administration, terminal elimination half-life is 4.5 hours.

Indication

X-ray contrast medium for use in adults and children for cardioangiography, arteriography, urography, phlebography and CT-enhancement. Lumbar, thoracic, cervical myelography and computed tomography of the basal cisterns, following subarachnoid injection. Arthrography, endoscopic retrograde pancreatography, (ERP), endoscopic retrograde cholangiopancreatography (ERCP), herniography, hysterosalpingography, sialography and studies of the gastrointestinal tract.

Recommended Dose

The Dosage vary depending on the type of examination, age, weight, cardiac output and general condition of the patient and the technique used. For intravenous, intra-arterial and intrathecal use, and use in body cavities.

Guidelines for Intravenous use

Indication	Concentration	Volume	Comments
Urography			
<u>adults:</u>	300 mg I/ml or 350 mg I/ml	40 - 80 ml 40 - 80 ml	80 ml may be exceeded in selected cases
<u>children < 7 kg</u>	240 mg I/ml or 300 mg I/ml	4 ml/kg 3 ml/kg	
<u>children > 7 kg</u>	240 mg I/ml or 300 mg I/ml	3 ml/kg 2 ml/kg (max 40 ml)	
Phlebography (leg)	200 mg I/ml or 240 mg I/ml or 300 mg I/ml	20 - 100 ml/leg	
Digital subtraction angiography	300 mg I/ml or 350 mg I/ml	20 - 60 ml/inj. 20 - 60 ml/inj.	
CT-enhancement	140 mg I/ml <u>adults:</u> or 200 mg I/ml or 240 mg I/ml or 300 mg I/ml or 350 mg I/ml	100 - 400 ml 100 - 300 ml 100 - 250 ml 100 - 200 ml 100 - 150 ml	Total amount of iodine usually 30 - 60 g
<u>Children:</u>	240 mg I/ml or 300 mg I/ml	2-3 ml/kgbw up to 40 ml 1-3 ml/kgbw up to 40 ml	In a few cases up to 100 ml may be given

Guidelines for Intrathecal use

Indication	Concentration	Volume	Comments
Lumbar and thoracic myelography (lumbar injection)	180 mg I/ml or 200 mg I/ml or 240 mg I/ml	10 - 15 ml 10 - 15 ml 8 - 12 ml	
Cervical myelography (lumbar injection)	240 mg I/ml or 300 mg I/ml	10 - 12 ml 7 - 10 ml	
Cervical myelography (lateral cervical injection)	240 mg I/ml or 300 mg I/ml	6 - 10 ml 6 - 8 ml	
CT cisternography (lumbar injection)	180 mg I/ml or 200 mg I/ml or 240 mg I/ml	5 - 15 ml 5 - 15 ml 4 - 12 ml	
Paediatric myelography			
<2 years	180 mg I/ml	2 - 6 ml	
2-6 years	180 mg I/ml	4 - 8 ml	
>6 years	180 mg I/ml	6 - 12 ml	
To minimize possible adverse reactions a total dose of 3 g iodine should not be exceeded.			

Guidelines for Intra-arterial use

Indication	Concentration	Volume	Comments
Arteriographies			
arch aortography	300 mg I/ml	30 - 40 ml/inj	Volume pr.
selective cerebral	300 mg I/ml	5 - 10 ml/inj	injection
aortography	350 mg I/ml	40 - 60 ml/inj	depends on the
femoral	300 mg I/ml or 350 mg I/ml	30 - 50 ml/inj	site of injection
various	300 mg I/ml	depending on type of examination	
Cardioangiography			
<u>adults:</u>			
left ventricle and aortic root inj.	350 mg I/ml	30 - 60 ml/inj.	
selective coronary arteriography	350 mg I/ml	4 - 8 ml/inj.	
<u>children:</u>	300 mg I/ml or 350 mg I/ml	depending on age, weight and pathology (max 8 ml/kg)	
Digital subtraction angiography	140 mg I/ml or 200 mg I/ml or 240 mg I/ml or 300 mg I/ml	1 - 15 ml/inj. 1 - 15 ml/inj. 1 - 15 ml/inj. 1 - 15 ml/inj.	depending on site of inj. occasionally large volumes - up to 30 ml - may be used

Guidelines for Body cavities

Indication	Concentration	Volume	Comments
Arthrography	200 mg I/ml or 240 mg I/ml or 300 mg I/ml or 350 mg I/ml	5 - 20 ml 5 - 20 ml 5 - 15 ml 5 - 10 ml	
ERP/ERCP	240 mg I/ml	20 - 50 ml	
Herniography	240 mg I/ml	50 ml	The dosage varies with the size of the hernia
Hysterosal-pingography	240 mg I/ml or 300 mg I/ml	15 - 50 ml 15 - 25 ml	
Sialography	240 mg I/ml or 300 mg I/ml	0.5 - 2 ml 0.5 - 2 ml	

Gastrointestinal studies

Oral use

Adults:

180 mg I/ml
or 200 mg I/ml
or 350 mg I/ml

individual
individual
individual

Children:

- esophagus

300 mg I/ml
or 350 mg I/ml

2-4 ml/kgbw
2-4 ml/kgbw

Max. dose 50 ml
Max. dose 50 ml

- ventricle/follow trough

140 mg I/ml

4-5 ml/kgbw

Prematures:

350 mg I/ml

2-4 ml/kgbw

Rectal use

- children:

140 mg I/ml
or dilute with tapwater to 100-150 mg I/ml

5-10 ml/kgbw
5-10 ml/kgbw

Example: Contrapaque 240, 300 or 350 with top-water 1:1 or 1:2

CT-enhancement

Oral use

- adults:

Dilute with tapwater to ~6mgI/ml

800 -2000 ml of the diluted solution over a period of time

Example: Contrapaque 300 or 350 with top-Water 1:50

- children:

Dilute with tapwater to ~6mgI/ml

15-20 ml/kgbw of the diluted solution

Rectal use

- children:

Dilute with tapwater to ~6mgI/ml

individual

Route of administration

For intravenous, intra-arterial and intrathecal use, and use in body cavities.

Contraindication

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Iohexol should not be administer to patients with known hypersensitivity to Iohexol.

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Iohexol should not be administered with intrathecal corticosteroids.

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Immediate repeat myelography with iohexol is contraindicated because of the possibility of overdosage.

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Myelography is contraindicated in the presence of significant local or systemic infection where bacteremia is likely.

SAP Code:142792

Country :Malaysia

Mfg. Location :Panoli IV17

Product Name : Contrapaque 300-350

Packaging Material Leaflet

Existing / Reference Art Party's Word File

Version No.: 1

Date : 03-11-2025

Dimension : 220 (L) x 240 mm (B) (± 1 mm tolerance)

Size After Folding :55 x 60 mm (± 1 mm tolerance)

Core Dia : NA

Reel Dia NA

Varnish / Lamination :NA

Roll Unwind Direction :NA

Print Repeat Length :NA

Specification : ITC Fine paper

Pantone :

Black

Folding pattern : 2 Vertical + 2 Horizontal Folds

Grammage: 40 ± 5% gm/m²

Thickness NA

Other Test :

Reason for Change New Artwork

Unique

J. B. CHEMICALS & PHARMACEUTICALS LTD.

File Path : X:\Artworks\Mahesh\Malaysia\Contrapaque V2\Open\ 142792 Contrapaque 300-350 Leaflet.ind

Artist : VIJAY

This document is equivalent to a digitally certified copy & digital signature is not required.

Front

Warning and precaution

General warning

- Patients should be well hydrated prior to and following Iohexol administration. Even invisible damage caused to the container during storage or transit may result in contamination. Before using check visually and do not use if precipitates or particulate matter observed or contents not clear or discoloured and return for replacement. The container once opened has to be used immediately. Any unused portion of contrast medium remaining in the container must be discarded.
- Patients with a previous reaction to a contrast medium, patients with known sensitivity to Iodine, or patients with clinical hypersensitivity (asthma, hay fever, food allergy) are at greater risk of developing serious, anaphylactic reactions. Pretesting does not delineate reliably the occurrence of severe reactions in these patients.
- A) Positive history of allergies or hypersensitivity does not preclude the use of a contrast agent if the procedure is considered essential; however, premedication with antihistamines or steroids to avoid or minimize allergic reactions should be considered. Although pretreatment may not prevent serious life-threatening reactions, it may reduce the incidence of severity of such reactions.
- B) Intravascular
Meticulous intravascular administration technique is required with both ionic and nonionic contrast media, particularly during angiographic procedures, to minimize thromboembolic events.
- C) Intrathecal
 - i. Caution is advised in patients with a history of seizures, severe cardiovascular disease, multiple sclerosis, or chronic alcoholism. Elderly patients may be at greater risk and the need for myelography should be carefully evaluated. Particular attention should be given to the dose and concentration of the medium, hydration, and technique used.
 - ii. Patients on anticonvulsant medication should be maintained on this therapy.
 - iii. Direct intracisternal or ventricular administration for standard radiography (without computerised tomographic enhancement) is not recommended. Also, effort should be directed to avoid rapid dispersion of the medium causing inadvertent rise of intracranial entry of medium occurs, prophylactic anticonvulsant treatment with diazepam or barbiturates orally for 24 to 48 hours should be considered.
 - iv. Use of medications that may lower the seizure threshold (phenothiazine derivatives, including those used for their antihistaminic or antinauseant properties; tricyclic antidepressants; MAO inhibitors; CNS stimulants; analeptics; antipsychotic agents; major tranquilizers) is not recommended. Such medication should be discontinued at least 48 hours before Iohexol administration, and should not be resumed for at least 24 hours post-procedure.

Thyroid dysfunction characterized by hypothyroidism or transient thyroid suppression has been reported in pediatric patients 0-3 years of age after exposed to iodinated contrast media. Younger age, very low birth weight, prematurity and other conditions are associated with an increased risk. If thyroid dysfunction is detected, treat and monitor thyroid function as clinically needed.

Take caution in patients with serious cardiac disease and pulmonary hypertension as haemodynamic changes or arrhythmias may developed.

Patients with acute cerebral pathology, tumours or a history of epilepsy are predisposed for seizures. Patients with paraproteinemias (myelomatosis and Waldenström's macroglobulinemia) are at risk of acute renal impairment

Interaction with other medicaments

- Drugs which lower seizure threshold, especially phenothiazine derivatives including those used for their antihistaminic or antinauseant properties, are not recommended for use with loxolol. Others include monoamine oxidase (MAO) inhibitors, tricyclic antidepressants, CNS stimulants, psychoactive drugs described as analeptics, major tranquilizers, or antipsychotic drugs. Such medications should be discontinued at least 48 hours before myelography, should not be used for the control of nausea or vomiting during or after myelography, and should not be resumed for at least 24 hours post procedure. In non-elective procedures in patients on these drugs, consider prophylactic use of anticonvulsants.
- Concurrent use of metformin is contraindicated in patients receiving intravascular iodinated contrast media. Injection of iodinated contrast materials in metformin-treated patients has been associated with lactic acidosis and can lead to acute renal failure. Metformin should be discontinued temporarily in patients undergoing radiologic studies involving intravascular administration of iodinated contrast materials. Withhold metformin prior to or at the time of the procedure and for 48 hours following the procedure. Restart metformin only after kidney function has been re-evaluated and found to be normal.
- Patient treated with interleukin-2 less than two weeks previously have been associated with an increased risk for delayed reactions (flu-like symptoms or skin reactions)

Pregnancy and lactation

- **Pregnancy:** Iohexol is classified as FDA pregnancy category B by the manufacturer, Iohexol should be used orally during pregnancy only if the potential benefit justifies the potential risk to the fetus. Animal studies performed on rats and rabbits have demonstrated no harm to the using up to 100times the recommended human dose Shaw and Potts, 1985). No studies have been performed in pregnant women.
- **Nursing mothers:** The amount of Iohexol excreted in breast milk has not been determined, but is probably small because of its high hydrophilicity. The use of Iohexol in nursing mothers should be made taking into account the importance of the drug to the mother.

Side effects

Intrathecal

- The most frequently reported adverse reactions with lohexol are headache, mild to moderate pain including backache, neckache and stiffness, nausea and vomiting. These reactions usually occur 1 to 10 hours after injection, and almost all occur within 24 hours. They are usually mild to moderate degree, lasting for a few hours, and usually disappearing within 24 hours. Rarely, headache is often accompanied by nausea and vomiting and tends to be more frequent and persistent in patients not optimally hydrated. Other reactions: Feeling of heaviness, hypotension, hypertension, sensation of heat, sweating, vertigo, loss of appetite, drowsiness, hypertension, photophobia, tinnitus, neuralgia, paresthesia, difficulty in micturition, and neurological changes. All were transient and mild with no clinical sequelae.

Intravascular

- Adverse reactions following the use of Iohexol are usually of mild to moderate severity. However, serious, life-threatening and fatal reactions, mostly of cardiovascular origin, have been associated with administration of Iodine-containing contrast media, including Iohexol. The injection of contrast media is frequently associated with the sensation of warmth and pain, especially in peripheral angiography; pain and warmth are less frequent and less severe with Iohexol than with many contrast media.

Cardiovascular system

- Vertigo (including dizziness and lightheadedness), pain, vision abnormalities (including blurred vision and photomas) headache, and taste perversion. Others including anxiety, fever, motor and speech dysfunction, convulsion, paresthesia, somnolence, stiff neck, hemiparesis, syncope, shivering, transient ischemic attack, cerebral infarction, and nystagmus.

Respiratory system

- Dyspnea, rhinitis, coughing, and laryngitis.

Gastrointestinal system

- Nausea and vomiting. Others including diarrhea, dyspepsia, cramp and dry mouth

Skin and appendages

Urticaria, purpura, abscess and pruritus.

Skin and Subcutaneous Tissue Disorders –

Severe cutaneous adverse reactions (e.g. Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS) and acute generalised exanthematous pustulosis (AGEP)) have been reported in post-marketing experience of iodinated contrast media. Post-marketing Experience - Endocrine disorders Frequency 'not known': hypothyroidism.

Anaphylactoid reactions may occur irrespective of the dose and mode of administration and mild symptoms of hypersensitivity may represent the first signs of a serious reaction. Discontinue administration immediately, and initiated specific therapy via the vascular access if necessary

Symptoms and treatment of overdose

Intrathecal

- Clinical consequences or overdosage with Iohexol have not been reported. However, based on experience with other nonionic myelographic media, physicians should be alert to a potential increase in frequency and severity of CNS-mediated reactions. Even use of a recommended dose can produce effects tantamount to overdosage, if incorrect management of the patient during or immediately following the procedure permits inadvertent early intracranial entry of a large portion of the medium. The intracisternal LD 50 value of Iohexol (in grams of iodine per kilogram body weight) is greater than 2.0 in mice

Intravascular

- Overdosage may occur. The adverse effects of overdosage are life-threatening and affect mainly the pulmonary and cardiovascular systems. The symptoms include: cyanosis, bradycardia, acidosis, pulmonary hemorrhage, convulsions, coma and cardiac arrest.
- Treatment of an overdosage is directed toward the support of all vital functions, and prompts institution of symptomatic therapy. The intravenous LD 50 values of Iohexol (in grams of Iodine per kg body weight) are 24.2 in mice and 15.0 in rats.

Renal function should be monitored for the next 3 days and if needed, haemodialysis may be used for clearance of excessive contrast medium. There is no specific antidote.

Storage condition

Store below 30°C. Protect from light and secondary X-ray.

Shelf life

4 years from the date of manufacture

Presentation

50 ml and 100 ml clear USP type I glass vials.



Manufactured in India by:
UNIQUE PHARMACEUTICAL LABORATORIES
 (A Div. of J.B. Chemicals and Pharmaceuticals Ltd)
 Plot No.4, Phase-IV, GIDC Industrial Area,
 Panoli-394 116, Dist.-Bharuch.
 *Trade Mark

Product Registration Holder and Imported by:

Unimed SDN BHD
53, Jalan Tembaga SD 5/2B,
Bandar Sri Damansara, 52200,
Kuala Lumpur, Malaysia

Revised on: April 2026

240 mm

220 mm